Ch 6.3: Dimension Reduction

Lecture 18 - CMSE 381

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Weds, Oct 26, 2022

Announcements

Last time:

- Shrinkage: Ridge and Lasso

This lecture:

- PCA / PCR
- ~~PLS~~

Announcements:

- Homework #6 due Friday

Next year:
rewrite PCR section
from book

Section 1

Last time

Shrinkage

Find β to minimize

$\sum (y_i - \hat{y}_i)^2$

$$RSS = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

L_1

subject to:

Least Squares:

No constraints

Ridge:

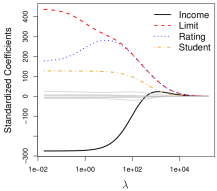
L_2

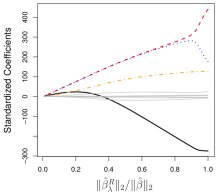
$\|\cdot\|_2$

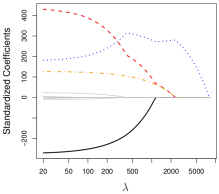
$$\sum_{j=1}^{p} \beta_j^2 \leq s$$

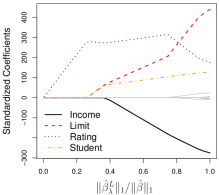
The Lasso:

$$\sum_{j=1}^{p} |\beta_j| \leq s$$









Section 2

Dimension Reduction

PCA

Linear transformation of predictors

Original Predictors:

$$X_1, \cdots, X_p$$

New Predictors:

$$Z_1, \cdots, \cancel{Z_M}\ Z_M$$

$$\underline{\underline{Z_m}} = \sum_{j=1}^{p} \varphi_{jm} X_j$$

R-value coefficient

old predictors

Goal: $M \ll p$

What choice of $\varphi = \{\varphi_{jm} \mid j = 1, \ldots, p,\ m = 1, \ldots, M\}$ does a good job?

An example or two

Input x_1, x_2

$$z_1 = x_1 + 3x_2$$

Input x_1, x_2, x_3

$$\varphi = \begin{pmatrix} \boxed{\begin{matrix} 1 \\ 0 \\ 0 \end{matrix}} & \begin{matrix} 0 \\ 1/2 \\ 1/2 \end{matrix} \end{pmatrix}$$

$$z_1 = 1 \cdot x_1 + 0 \cdot x_2 + 0 \cdot x_3$$

$$z_2 = 0 \cdot x_1 + \frac{1}{2} x_2 + \frac{1}{2} x_3$$

The goal

$X_1, \ldots, X_p$ predict Y

$\Downarrow$

$Z_1, \ldots, Z_M$ predictors

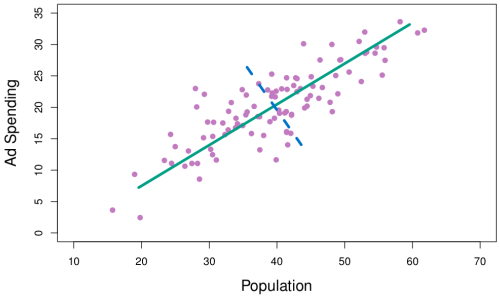
- Find good φ's
- Fit regression model on Z_i's using least squares

$$y_i = \theta_0 + \sum_{m=1}^{M} \theta_m \underline{\underline{z_{im}}} + \varepsilon_i$$

Section 3

PCA

An example dataset



By eye: Green line is the direction of maximum variability

Called the principal component

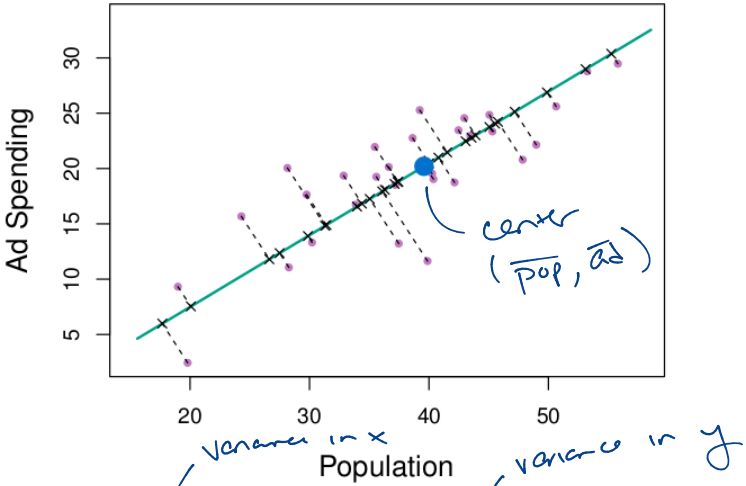
Translation:

If we project the data onto the principal component the projected data has max possible variance.

Projection onto a line

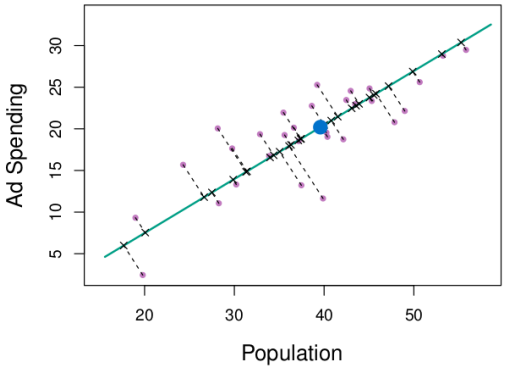
https://www.desmos.com/calculator/cih7wy8oyg

Projection onto first PC



$$Z_1 = 0.839 \cdot (\text{pop} - \overline{\text{pop}}) + 0.544 \cdot (\text{ad} - \overline{\text{ad}})$$

What does it mean to have the highest variance



Of any linear combination of pop and ad, this green line has highest variance

Maximizes

$$Var\left(\varphi_{11}(var - \bar{var}) + \varphi_{21}(ad - \bar{ad})\right)$$

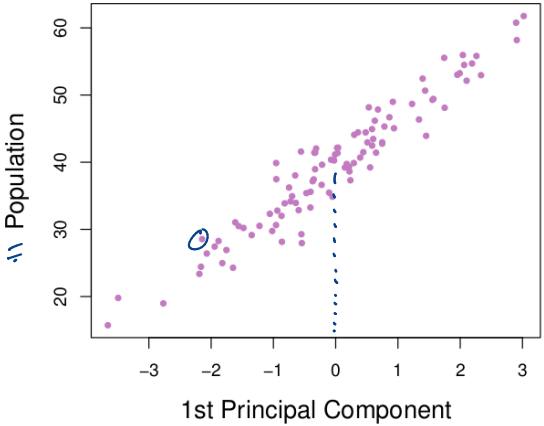
Require $\varphi_{11}^2 + \varphi_{21}^2 = 1$

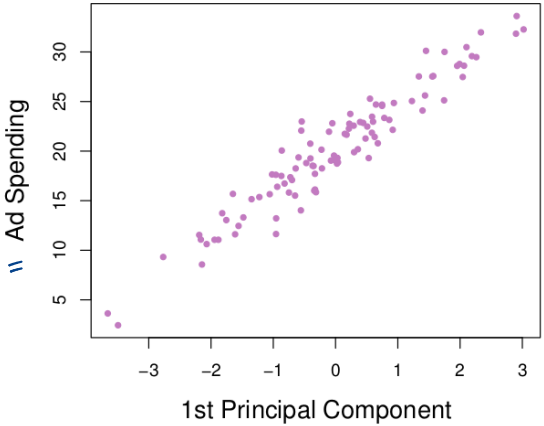
(else just make large to increase variance)

Toy for learning PCA

https://www.desmos.com/calculator/qq14tyjz0z

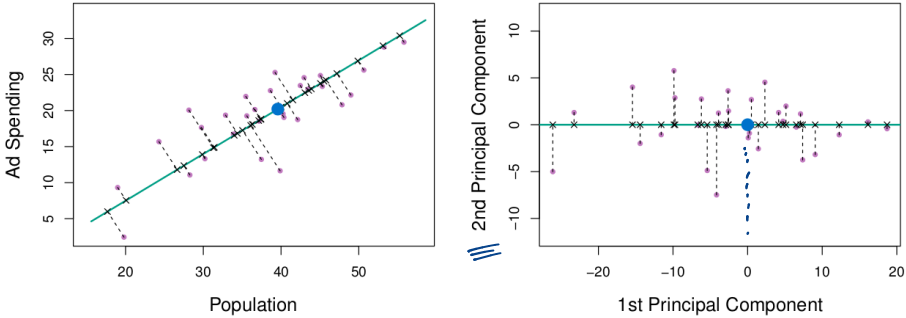
Principal component scores





$$z_{i1} = 0.839 \cdot (\text{pop}_i - \overline{\text{pop}}) + 0.544 \cdot (\text{ad}_i - \overline{\text{ad}})$$

Another view



Score can can be thought of as dist from mean

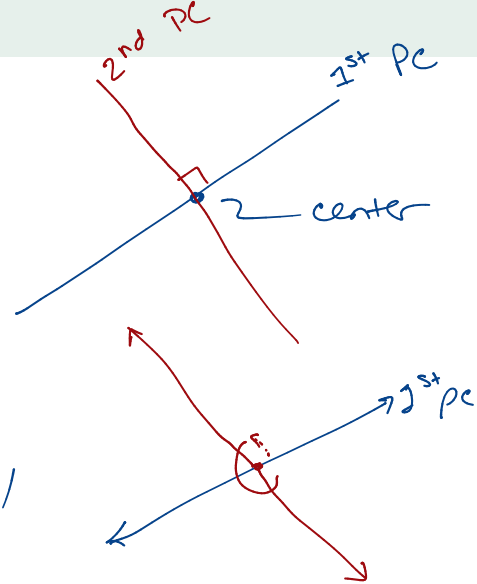
The other principal components

You have z_1

z_2 will be a linear comb of variables uncorrelated to z_1

Find the one that "explains" the most variance

Result: z_2 will be perpendicular / orthogonal to z_1



Do PCA with Penguins

Section 4

Principal Components Regression

PCA + Regression

So you've found your PCA coefficients

Now what?

$X_1, \ldots, X_p$ $\quad$ Y

not used for PCA

$\Downarrow$

$Z_1, \ldots, Z_M$ $\quad$ Y

Do linear regression on the Z's

$$Y = \beta_0 + \beta_1 Z_1 + \beta_2 Z_2 + \cdots + \beta_M Z_M$$

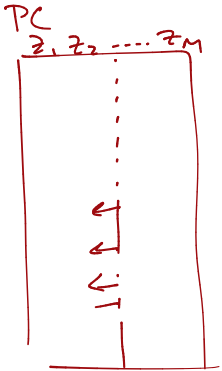
What are we assuming?

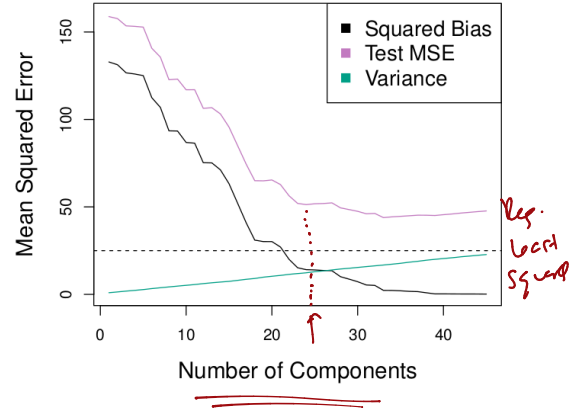
- The directions in which $X_1, \ldots, X_p$ have the most variance are the directions associated with Y
- If this is true, then we will get better predictions using PC's because less chance of overfitting

Doing better

Example with simulated data: $n = 50$ observations of $p = 45$ predictors

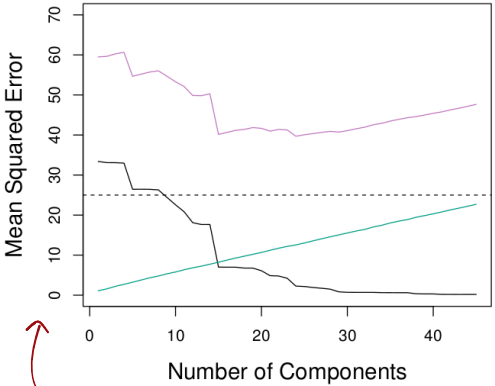
Y is a function of 2 predictors





Doing better

Example with simulated data: $n = 50$ observations of $p = 45$ predictors
Y is a function of all predictors



least squares

using PC's is a bigger improvement over least squares

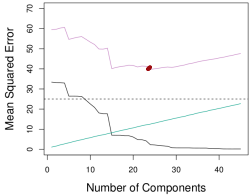
much smaller values on y-axis

Comparison to results on shrinkage

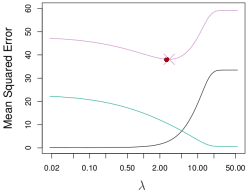
Possibly backwards pics?

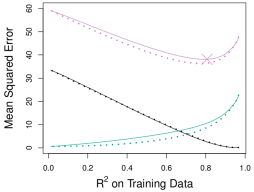
Y is a function of all predictors

6.18a



PCR



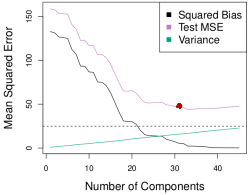


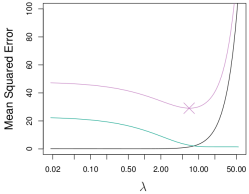
Ridge Lasso

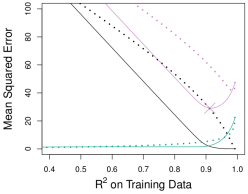
Lasso + ridge

Y is a function of 2 predictors

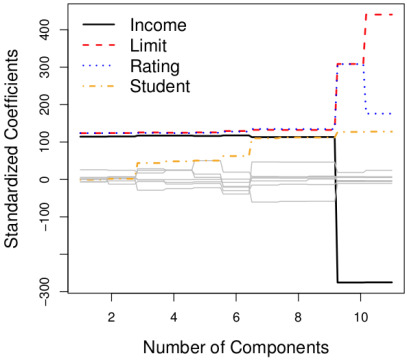
6.18b

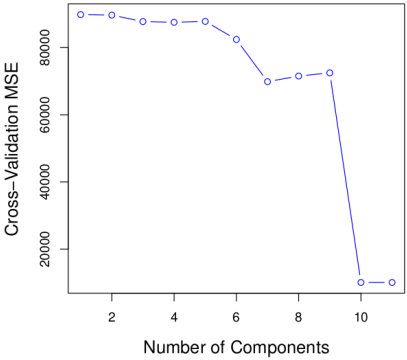






Picking M


Income
Limit
Rating
Student
Standardized Coefficients
−300
−100
0
100
200
300
400
2
4
6
8
10
Number of Components


Cross−Validation MSE
20000
40000
60000
80000
2
4
6
8
10
Number of Components

Properties of PCR

- More useful when info is contained in only a few PC's
  - ↳ when doing dimension reduction is helpful
- Usual tricks to decide on # of PC's using CV
- Want to standardize the data first so that large values don't skew the results
- Warning: Can't use the coefficients to make claims about the original variables

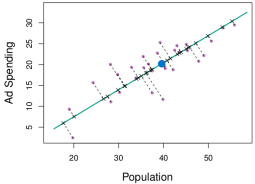
Do PCR with hitters data

- PCA ⇝ get fewer variables
- Run lin reg to do prediction

TL;DR

PCA

- Unsupervised dimensionality reduction
- Choose component Z_1 in the direction of most variance using only X_i's information
- Choose Z_2 and beyond by the same method after "getting rid" of info in the directions already explained



PCR

- Do PCA on input data
- Do Linear Regression on chosen number of PCs.
- Warning: Lose interpretability of the coefficients.

Next time

10	M	Oct 3	Leave one out CV	5.1.1, 5.1.2	
11	W	Oct 5	k-fold CV	5.1.3	
12	F	Oct 7	More k-fold CV,	5.1.4-5	
13	M	Oct 10	k-fold CV for classification	5.1.5	HW #4 Due
14	W	Oct 12	Resampling methods: Bootstrap	5.2	
15	F	Oct 14	Subset selection	6.1	
16	M	Oct 17	Shrinkage: Ridge	6.2.1	HW #5 Due
17	W	Oct 19	Shrinkage: Lasso	6.2.2	
18	F	Oct 21	[No class, Dr Munch out of town]		
	M	Oct 24	No class - Fall break		
19	W	Oct 26	Dimension Reduction	6.3	
20	F	Oct 28	More dimension reduction; High dimensions	6.4	HW #6 Due
	M	Oct 31	Review		
	W	Nov 2	***Midterm #2***		